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midterm #2

This is the most confused I have ever been & I am very disappointed in myself

I am so sorry

This was very hard

much harder than midterm #1

not an study guide?

#1

$f_{max} = \frac{1}{\text{total delay}}$

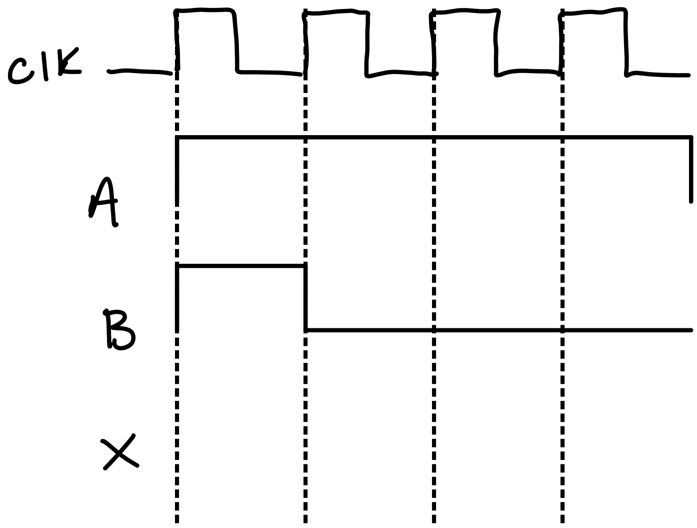
a) $f_{max} = \frac{1}{32ns} = 0.03125 \times 10^9$ HZ

$f_{max} = 31.25$ MHZ

b) total delay = 64ns

$f_{max} = \frac{1}{64ns} =$ 15.625 MHZ

c) timing diagram

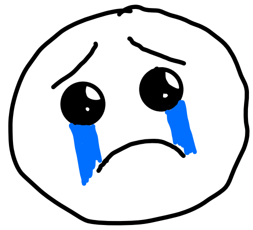


A = 1111 B = 1000

$$\begin{array}{r} 1111 \\ +\,1000 \\ \hline 10111 \end{array}$$

??

please take pity



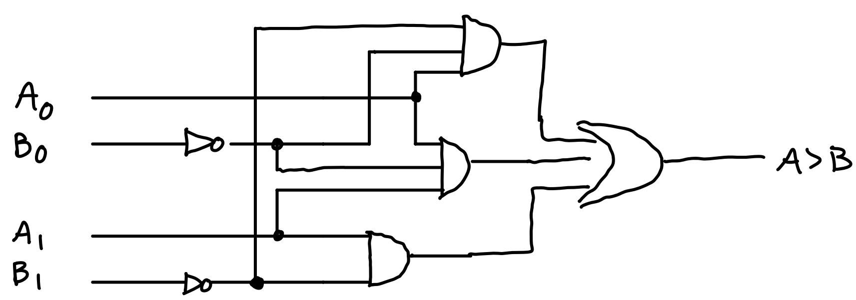
#2

sorry, my pycharm wasn't working & I couldn't use waveDrom

a) create 2-bit comparator that yeilds TRUE if A > B

A[1] A[0] \ B[1] B[0]	00	01	11	10
00	0	0	0	0
01	1	0	0	0
11	1	1	0	1
10	1	1	0	0

$A_0\bar{B}_1\bar{B}_0 + A_1\bar{B}_1 + A_1A_0\bar{B}_0$



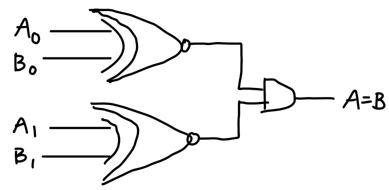
b) TRUE if A = B

A[1] A[0] \ B[1] B[0]	0 0	0 1	1 1	1 0
0 0	1	0	0	0
0 1	0	1	0	0
1 1	0	0	1	0
1 0	0	0	0	1

$$\overline{A_0 A_1 B_0 B_1} + A_1 A_0 B_1 B_0 + \bar{A}_1 A_0 \bar{B}_1 B_0 + A_1 \bar{A}_0 B_1 \bar{B}_0$$

$$= \overline{A_1 B_1}\left(\overline{A_0 B_0} + A_0 B_0\right) + A_1 B_1\left(A_0 B_0 + \overline{A_0 B_0}\right)$$

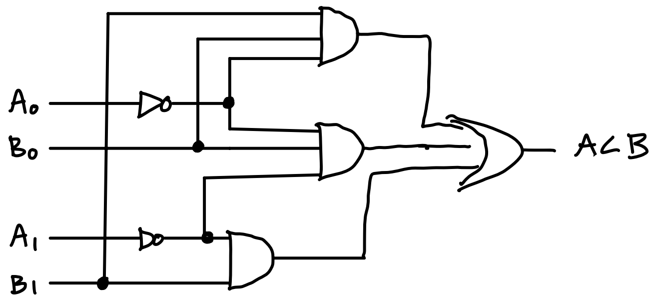
$$= (A_0 \odot B_0)(A_1 \odot B_1)$$



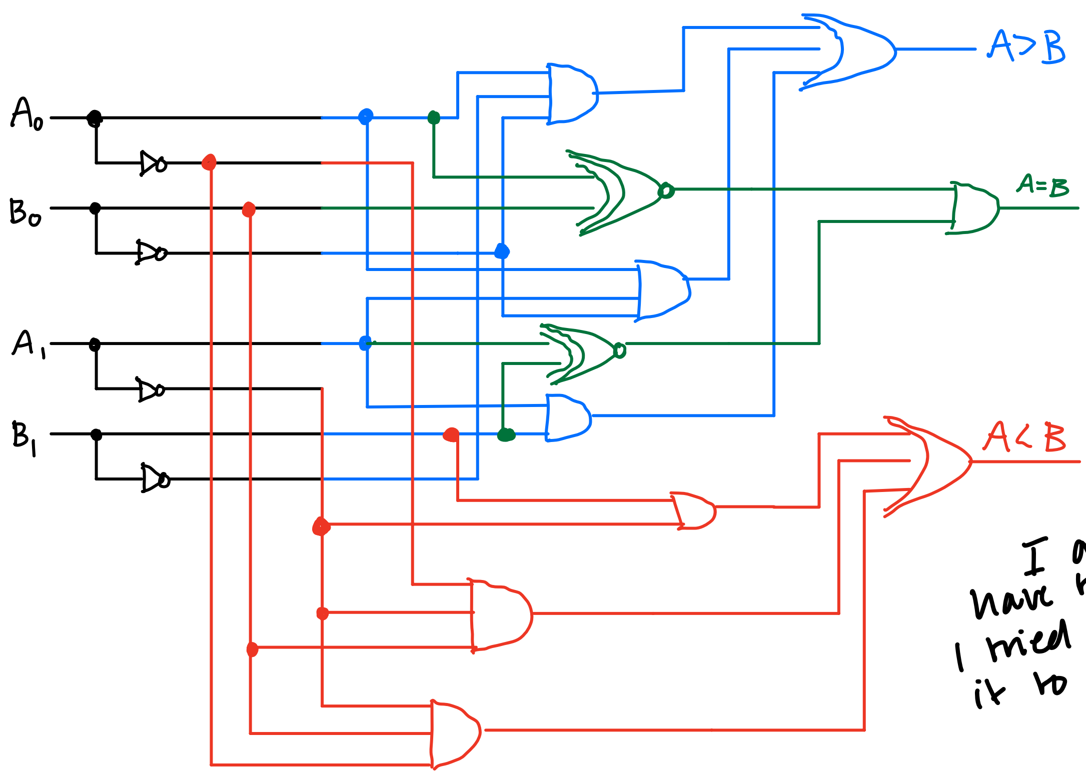
c) TRUE if A < B

A[1] A[0] \ B[1] B[0]	0 0	0 1	1 1	1 0
0 0	0	1	1	1
0 1	0	0	1	1
1 1	0	0	0	0
1 0	0	0	1	0

$$= \overline{A_1}\,\overline{A_0} B_0 + \bar{A}_1 B_1 + \bar{A}_0 B_1 B_0$$



d) combine into one circuit w/ 3 outputs & 4 inputs



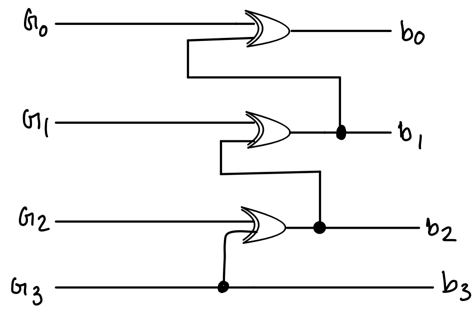
I am sorry you have to look at this
I tried to color code it to make it easier

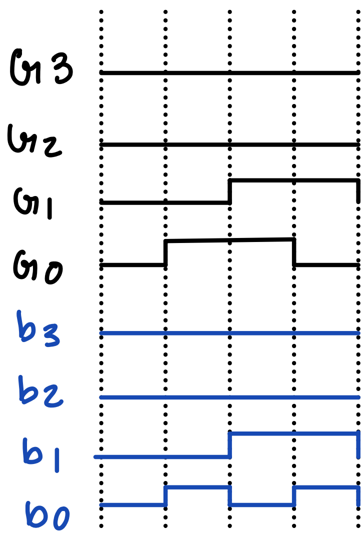
#3 Generate the logic gates that convert 4-bit gray code to 4-bit binary code, and show that it works w/a timing diagram:

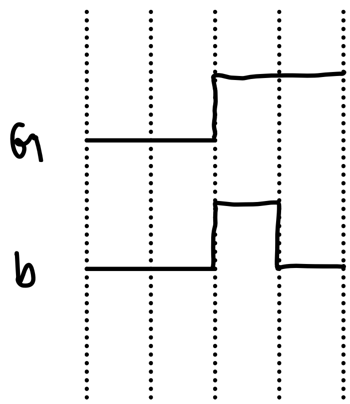
how :(

Gray ⇒ Binary

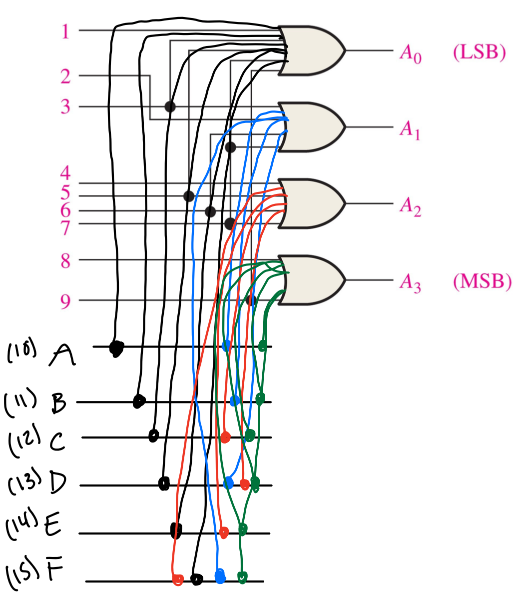
0000	0000
0001	0001
0011	0010
0010	0011







#4 a) add logic to the circuit such that it becomes a hex to binary converter



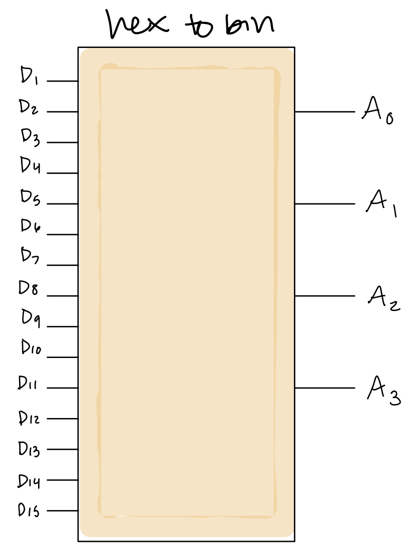
added black lines go to A_0

added blue lines go to A_1

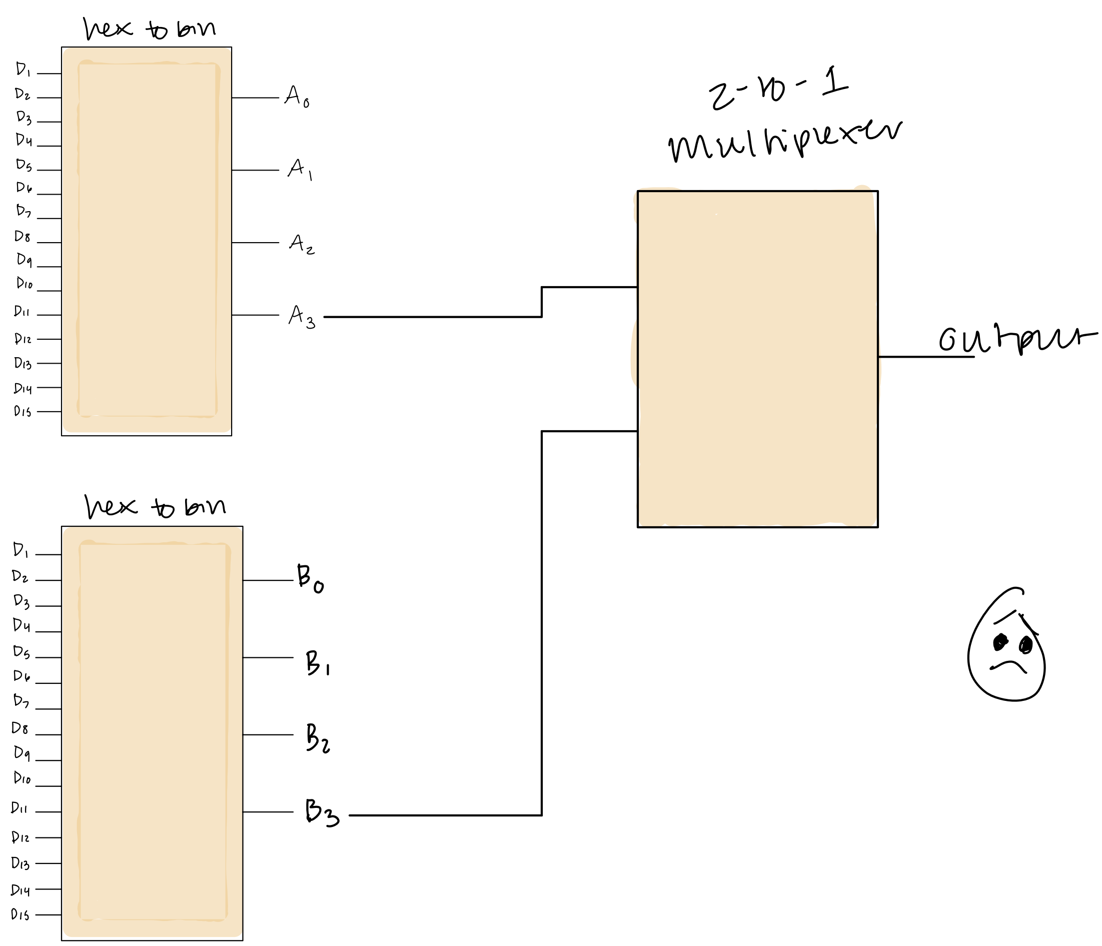
added red lines go to A_2

added green lines go to A_3

b) draw logic symbol for this circuit w/ the correct number of inputs + outputs

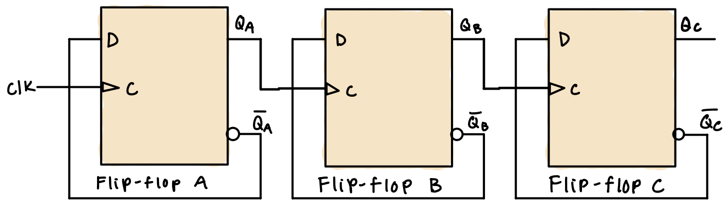


c) connect two hex-to-bin converters to a symbolic 2-to-1 multiplexer . . .

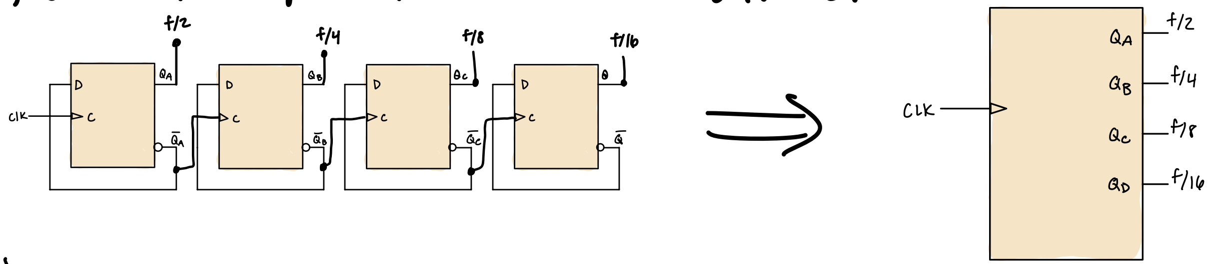


Ch. 7

#1 a) elaborate on this circuit to create a circuit that can divide the clock frequency by 2, 4, or 8. Show flip-flops explicitly

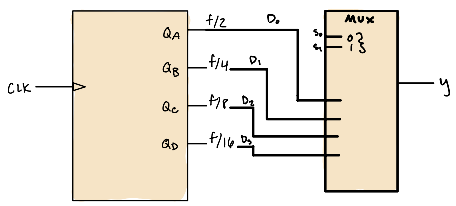


b) develop symbol for 2-4-8-16 divider



?

c)



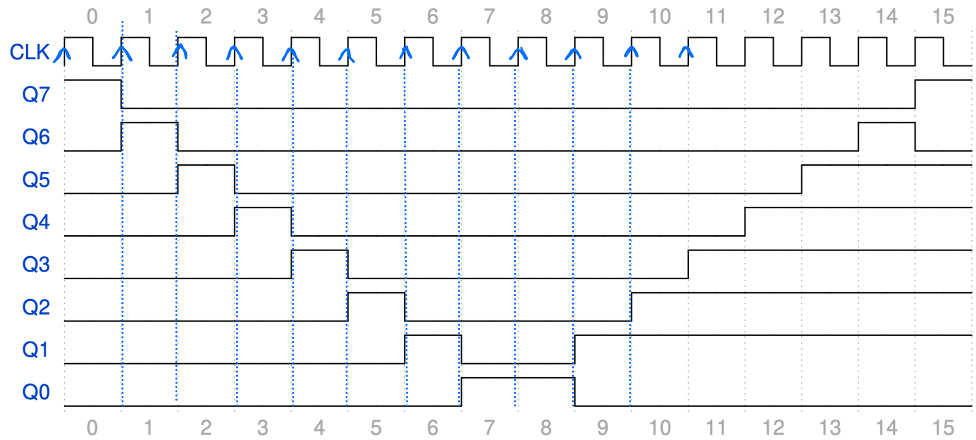
d) show timing diagram the output ??
if user changes data select lines at least once

what

ch 8+9

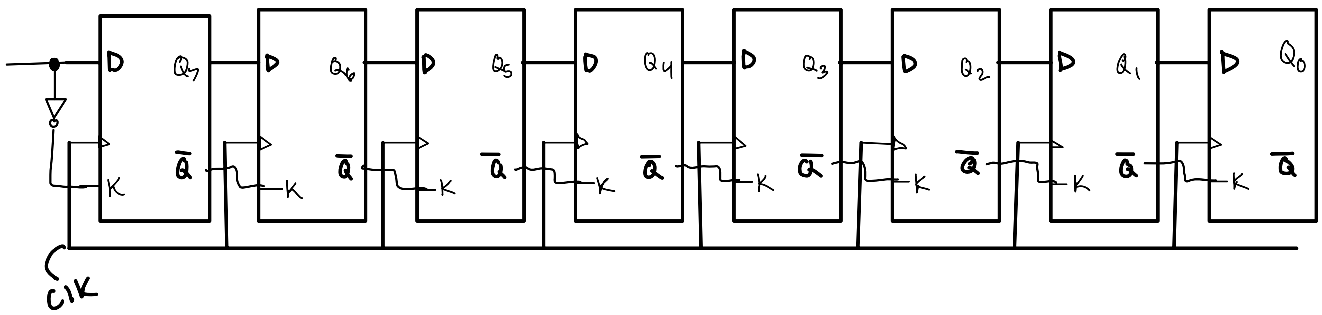
#1 using any combination of shift-registers + supporting gates, create circuit w/ 8 outputs that produces timing diagram

pulse clk #	Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0
0	1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0
3	0	0	0	1	0	0	0	0
4	0	0	0	0	1	0	0	0
5	0	0	0	0	0	1	0	0
6	0	0	0	0	0	0	1	0
7	0	0	0	0	0	0	0	1
8	0	0	0	0	0	0	0	1
9	0	0	0	0	0	0	1	0
10	0	0	0	0	0	1	1	0
11	0	0	0	0	1	1	1	0
12	0	0	0	1	1	1	1	0
13	0	0	1	1	1	1	1	0
14	0	1	1	1	1	1	1	0
15	1	0	1	1	1	1	1	0



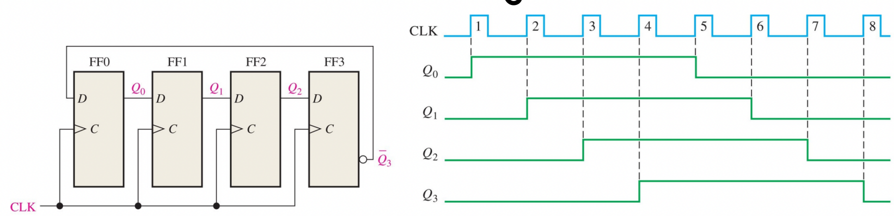
rising edge signal

very confused on how to do this one

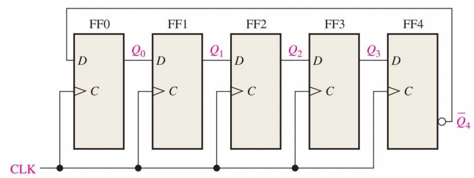


#2 consider fig 4 in which 4 and 5-bit johnson counters are depicted

a) create a circuit based on fig. 4 called a combinatoric trigger. Imagine four digital channels A, B, C, D as inputs. Output of circuit should be true if any two of the four channels is true w/in 100 ns of eachother. Develop any necessary logic symbols & use appropriate clock frequency. The johnson counter(s) can have any number of bits



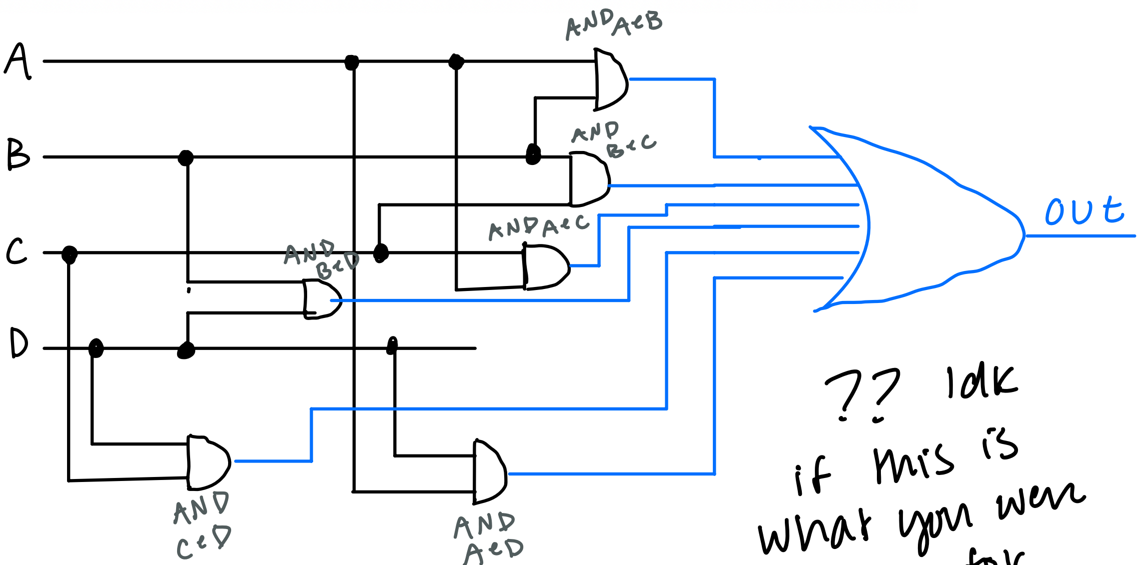
(a) Four-bit Johnson counter



(b) Five-bit Johnson counter

Need clock frequency that ensures that cycle corresponds to time window

$$\text{clock frequency} = \frac{1}{100\,ns} \Rightarrow 10\ MHz$$



?? idk if this is what you were asking for

each input is AND w/ eachother to see if true w/in 100 ns

↳ signals trigger OR gate if combination is detected

{ch. 12}

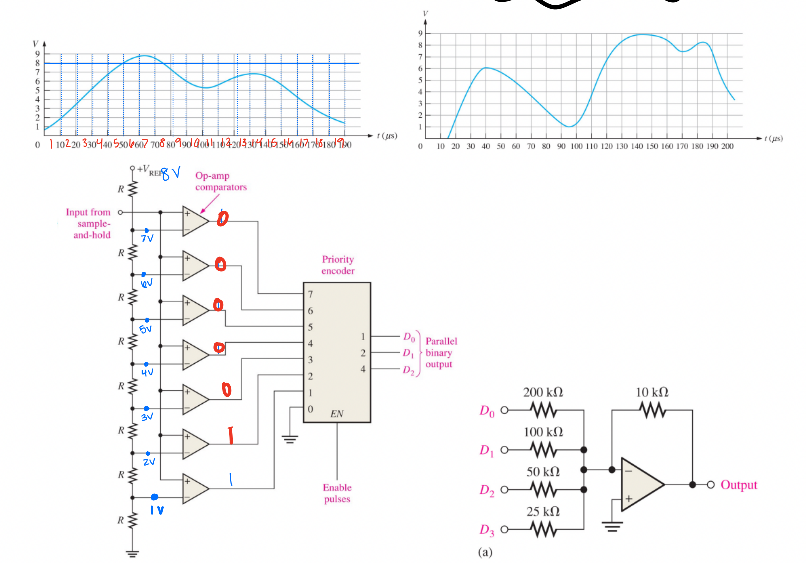
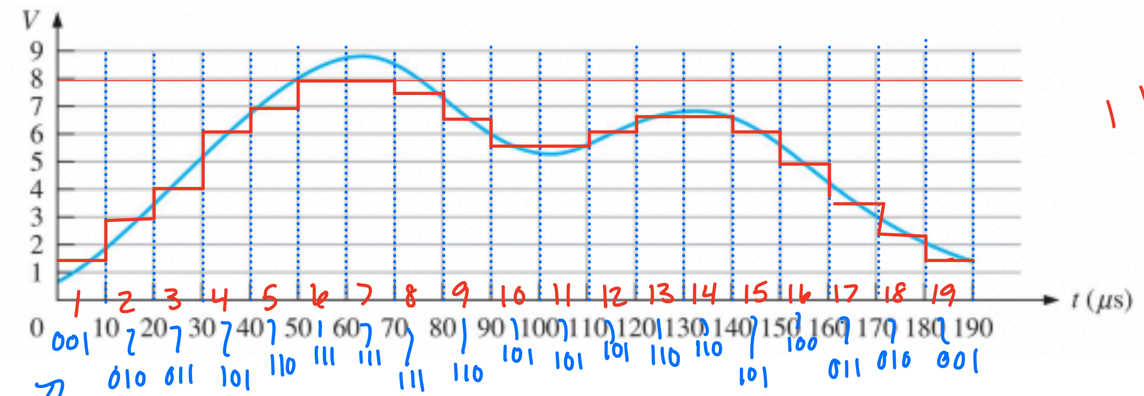


Figure 5: (Top left) An input waveform for a 3-bit flash ADC with V_{CC} = 8 Volts. (Top right) A second input waveform for a 3-bit flash ADC with V_{ref} = 8 Volts. (Bottom left) The 3-bit flash ADC. (Bottom right) A 4-bit DAC with binary weighted inputs.

#1 a) determine the binary output codes from the 3-bit flash ADC if fig. 5 given the input waveform

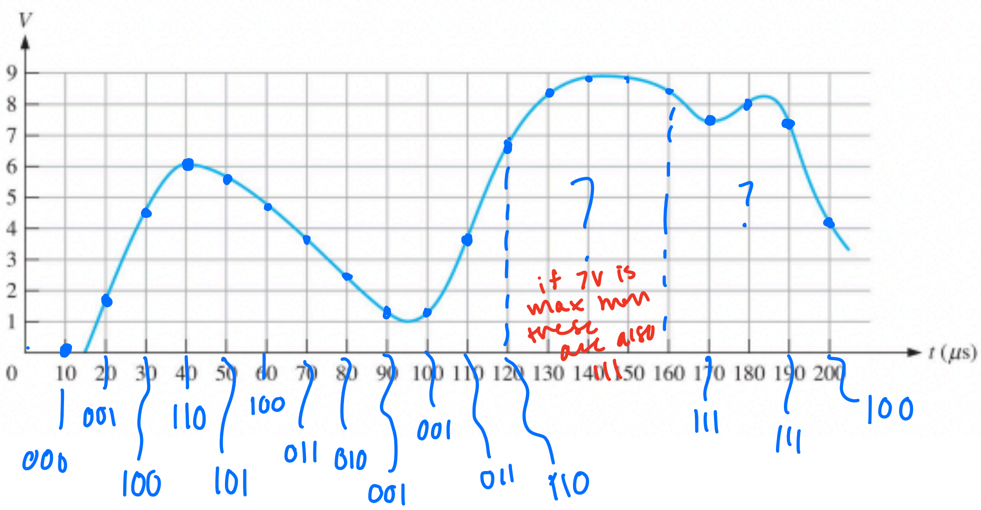


binary output codes

I think?

if Vcc > v, then it goes to priority encoder which will tell us the binary output codes

b) repeat but for the other input waveform

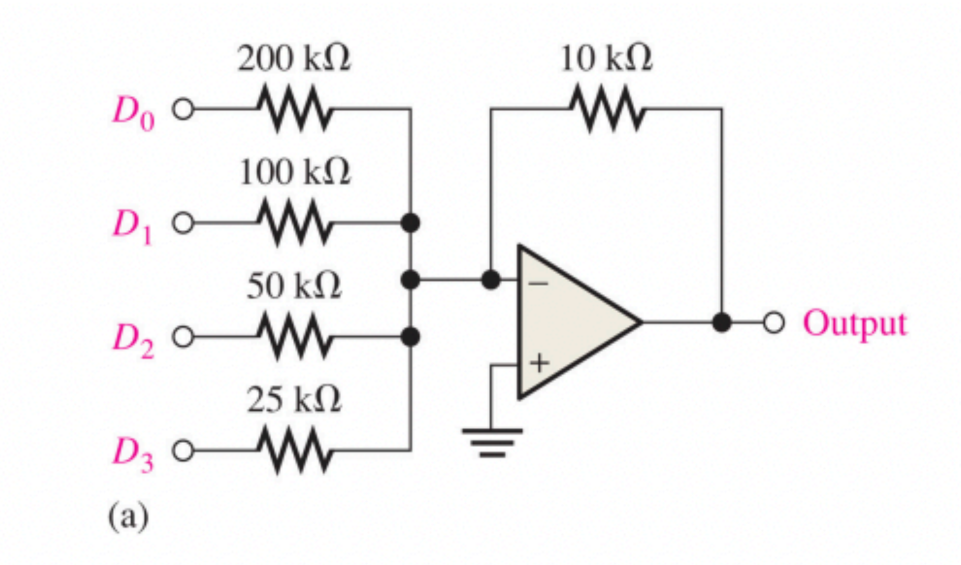


I did these ones different than in part (a)

I think max V it can be is 7 volts if V_{ref} = 8 V

8/8 = 1V drop between resistors

#2 Determine the outputs of the DAC if the sequence of 4-bit numbers is applied to the inputs. The data values have a low value of 0V and high of 5V



$I_0 = \frac{5V}{200\,k\Omega} = 0.025\ mA$ $V_{out}(D_0) = (10\,k\Omega)(-0.025\,mA) = -0.25V$

$I_1 = \frac{5V}{100\,k\Omega} = 0.05\ mA$ $V_{out}(D_1) = (10\,k\Omega)(-0.05\,mA) = -0.5V$

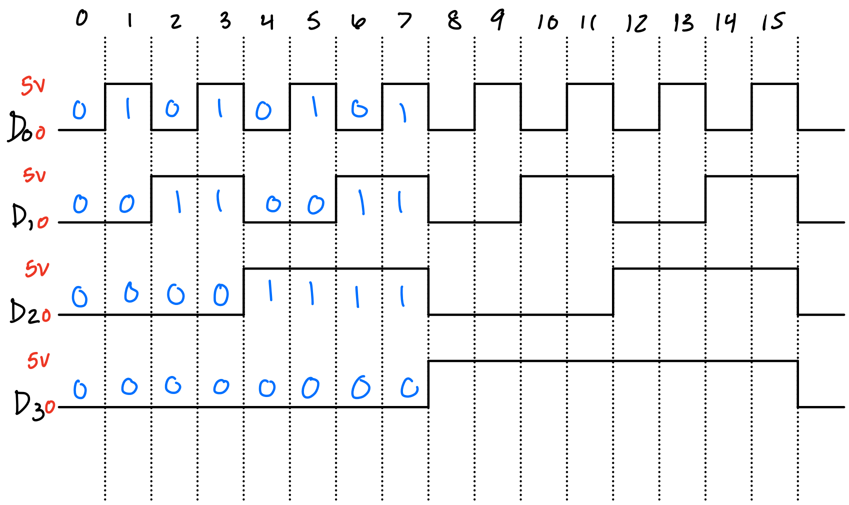
$I_2 = \frac{5V}{50\,k\Omega} = 0.1\ mA$ $V_{out}(D_2) = (10\,k\Omega)(-0.1\,mA) = -1V$

$I_3 = \frac{5V}{25\,k\Omega} = 0.2\ mA$ $V_{out}(D_3) = (10\,k\Omega)(-0.2\,mA) = -2V$

is negative b/c $V_{out} = -IR$

Sequence of 4-bit numbers not given??

→ will assume sequence is binary 1-15



0000 ⇒ $V_{out} = 0V$

0001 ⇒ $V_{out} = -0.25V$

0010 ⇒ $V_{out} = -0.5V$

0011 ⇒ $V_{out} \Rightarrow -0.25 + -0.5 = -0.75V$

0100 ⇒ $V_{out} = -1V$

0101 ⇒ $V_{out} = -1V + -0.25 = -1.25V$

0110 ⇒ $V_{out} = -1V + -0.5V = -1.5V$

⋮ and so on

each bit in the binary number has a V_{out} (calculated above), for each binary number you must add them up to get the total V_{out} for that number